

Visual System Reset Report

This pass creates a stricter visual-aid pattern and rebuilds RAG and Retrieval as the gold-standard Open-Book Retrieval visual. It does not start Batch 3, add games, or add 3D libraries.

What Changed

- Added the visual pattern: `VisualAidCard`, `DiagramScene`, `CalloutList`, and `KeyTakeaway`.
- Added learning objective, accessible description, key takeaway, and print/PDF notes to visual metadata.
- Rebuilt RAG as a four-zone diagram: Prompt, Retriever, Notes, Context, Response, plus fixed weights.
- Completed RAG lesson fields for durable-vs-temporary, prompt-vs-response, core explanation, misconceptions, and feedback.
- Improved the visual gallery so review cards show visual id, lesson, scene, callouts, accessible description, mobile preview, and print/PDF notes.
- Updated `Open Book or Learned?` with the requested RAG sorting items.
- Bumped the visible app version to `v0.9.3` and updated export filenames.

RAG Finding

RAG was current in the repo but visually unrepaired. The source already had the `rag-retrieval` Journey card, but it still used an older mini-infographic visual and thinner lesson fields because v0.9.2 focused mostly on Batch 1 and Batch 2 visuals.

Files Changed

- `README.md`, `package.json`, and `scripts/export-lesson-pdf.mjs`
- `src/components/VisualAids.tsx`
- `src/data/content.ts`, `src/data/contentReview.js`, and `src/data/exercises.ts`
- `src/main.tsx` and `src/styles/global.css`
- `docs/REVIEW_NOTES.md`
- v0.9.3 curriculum docs, PDFs, screenshots, and this report

Verification

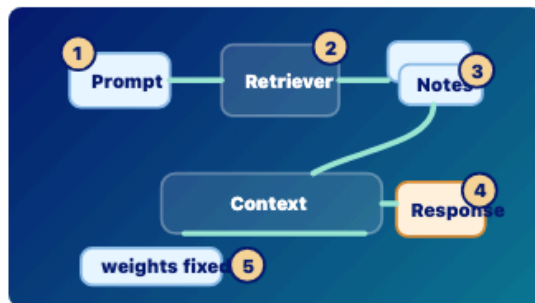
- `npm install`: passed, 0 vulnerabilities.
- `npm run typecheck`: passed.
- `npm run build`: passed.
- `npm run build:pages`: passed.
- `npm run export:lesson-pdf`: passed.
- `npm run export:lesson-cards`: passed.
- In-app browser QA confirmed the RAG visual and completed RAG fields.
- 320px, 390px, and 430px checks showed no horizontal overflow for RAG, Training Loop, Prompt vs Response, visual gallery, or the RAG lesson-card sample.

Screenshots

Open-Book Retrieval

Lesson: RAG and Retrieval

Pattern: rag



Open-Book Retrieval

Retrieval plus context, not training

Retrieved notes enter the context before response tokens are generated.

- 1 **Ask** The user prompt starts the run.
- 2 **Retrieve** A search system finds relevant outside material.
- 3 **Add to context** Retrieved notes become temporary context tokens.
- 4 **Generate** The model still generates response tokens one at a time.
- 5 **Weights stay fixed** RAG does not normally update model weights.

Key takeaway:

RAG is retrieval plus context, not training.

LEARNING OBJECTIVE

Show that RAG retrieves outside information and places it into context; it does not train the model.

ACCESSIBLE DESCRIPTION

The RAG diagram moves from Prompt to Retriever to Notes, then into a Context tray and Generated response, with a separate fixed-weights note.

MOBILE PREVIEW

Review at 320px, 390px, and 430px; diagram labels must stay inside the canvas.

PRINT/PDF NOTES

Gold-standard v0.9.3 visual: short diagram labels, five HTML callouts, and a one-sentence takeaway.

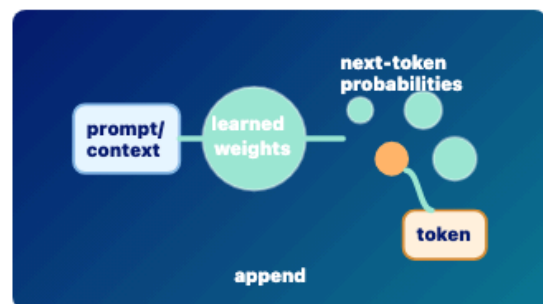
Visual Aid Gallery

Reusable SVG/CSS diagrams for the current lesson path, with mobile and PDF readability checks.

Prompt to Prediction

Lesson: What Is an LLM?

Pattern: llmOverview



Prompt to Prediction

Prompt/context flows through learned weights into next-token probabilities; one generated response token is appended.

- 1 **Point 1** Prompt/context is given input.
- 2 **Point 2** Learned weights are used during inference.
- 3 **Point 3** The selected response token is appended.

LEARNING OBJECTIVE

Prompt/context flows through learned weights into next-token probabilities; one generated response token is appended.

ACCESSIBLE DESCRIPTION

Prompt/context flows through learned weights into next-token probabilities; one generated response token is appended.

MOBILE PREVIEW

Review at 320px, 390px, and 430px; diagram labels must stay inside the canvas.

PRINT/PDF NOTES

Use the same scene and HTML callouts in exported lesson-card PDFs.

Learning Modes

Lesson: How AI Learns Pattern: learns



RAG gold standard: retrieval plus context, not training.

TRADITIONS

Rules and Learned Patterns

Visual gallery with review metadata.

TRAINING-LOOP

Training Loop

Lesson: Training

Pattern: training



Training Loop

Durable change happens at weight update

Predict, compare, loss, update weights, repeat. Training changes weights.

- 1 Predict** The model predicts a target.
- 2 Compare** The prediction is compared with the target.
- 3 Loss** Loss measures error.
- 4 Update weights** Weight updates are the durable-change step.
- 5 Repeat** The loop repeats many times.

Key takeaway:

Training changes weights; ordinary inference does not.

LEARNING OBJECTIVE

Show the sequence of training and make the durable weight-update step visually distinct.

ACCESSIBLE DESCRIPTION

A five-step loop moves from Predict to Compare to Loss to Update weights to Repeat, with Update weights highlighted.

MOBILE PREVIEW

Review at 320px, 390px, and 430px; diagram labels must stay inside the canvas.

PRINT/PDF NOTES

Five nodes stay aligned at 320px and in exported review PDFs.

PRETRAINING-RAIN

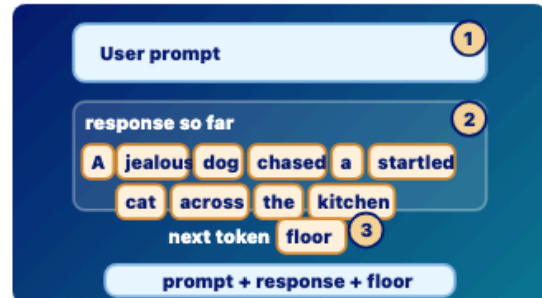
Broad Pretraining

PROMPT-RESPONSE

Prompt vs Response

Lesson: Prompt vs Response

Pattern: promptResponse



Prompt vs Response

Given context versus generated tokens

A complete user prompt is given; response tokens are generated and appended.

- 1 User prompt** The complete request is: Give me one sentence with two household pets in conflict.
- 2 Response so far** The model has already generated: A jealous dog chased a startled cat across the kitchen.
- 3 Next token** floor is appended, so the next context combines prompt plus response so far plus floor.

Key takeaway:

The response grows one token at a time inside the current context.

LEARNING OBJECTIVE

Separate the complete user prompt from the incomplete response-so-far and next generated token.

ACCESSIBLE DESCRIPTION

The visual separates a complete user prompt, response-so-far token chips, the next token floor, and the next context row.

MOBILE PREVIEW

Review at 320px, 390px, and 430px; diagram labels must stay inside the canvas.

PRINT/PDF NOTES

Token chips wrap inside the visual frame; full example text stays in HTML callouts.

TOKENIZATION

Lesson: Pretraining

Pattern: pretraining



Training Loop with a widened Update weights node.

Text to Tokens

Lesson: Tokenization

Pattern: token



Prompt vs Response using the v0.9.2 canonical example.

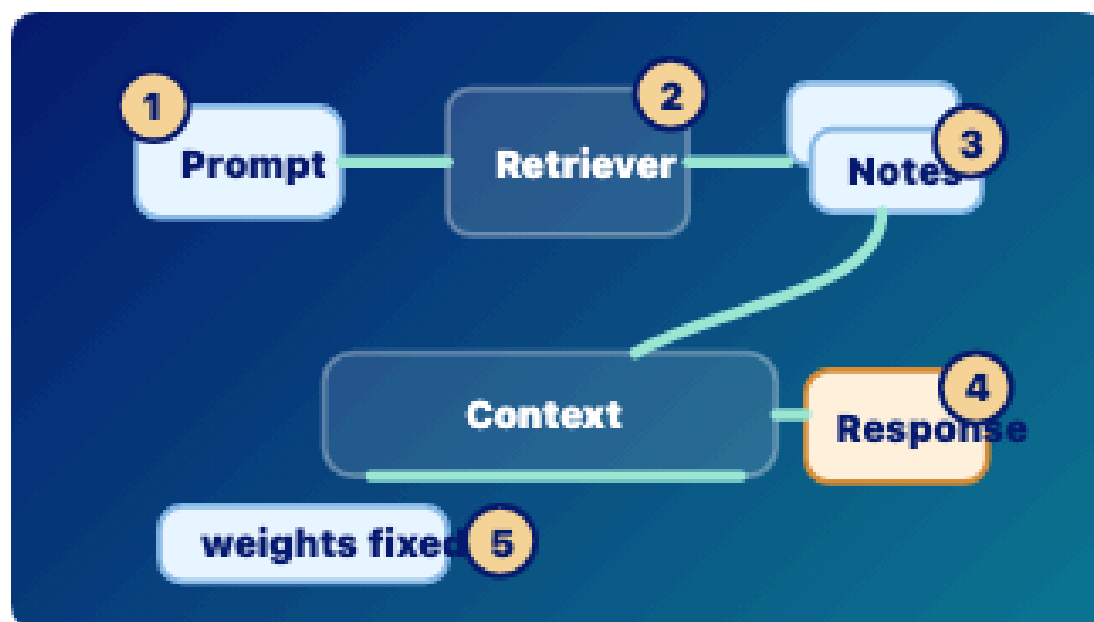
RAG and Retrieval

Open-book AI

Stage: retrieval-inference

Priority: high

Rubric avg: 4.6/5



Open-Book Retrieval

Retrieval plus context, not training

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- temporary context tokens.
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Key takeaway:

RAG is retrieval plus context, not training.

DEFINITION

Retrieval-augmented generation, or RAG, retrieves outside information and places it into the model's current context before the model generates a response.

CORE EXPLANATION

A plain model answers from learned weights and the current context. A RAG system adds a retrieval step. It searches documents, passages, or records, then inserts relevant snippets into the current context. The model then generates a response using its learned weights plus the retrieved context.

WHERE IT HAPPENS

During inference, before or during

during inference, before or during response generation.

DURABLE VS TEMPORARY

RAG usually does not change model weights. Retrieved information is temporary context unless a separate

RAG lesson-card sample from the print route used by PDF export.

Known Remaining Issues

- Some older visuals outside the named repair targets still use the legacy metadata fallback and should be converted gradually.
- Source citations are still needed before publication-final claims about RAG and transformer internals.
- Real iPhone Safari testing remains the final check after GitHub Pages deploy, especially for browser cache and safe-area behavior.

Ready For Batch 3?

Yes for visual-system foundation after final deploy verification. Batch 3 should use the RAG pattern as the model: one objective, simple scene, numbered HTML callouts, key takeaway, and accessible description.

Next Three Steps

- Build Batch 3 for Attention, MLP, Layers, and Hidden States using this stricter visual pattern.
- Add source citations or bibliography notes for RAG and transformer internals.
- Convert remaining legacy visuals to the card/scene/callout/takeaway pattern over time.

How To Run

- `cd ~/Desktop/promptlife`
- `npm install`
- `npm run dev`
- Open `http://localhost:5173/`.
- For iPhone testing after deploy, open `https://kevinhegg.github.io/promptlife/?v=093` or confirm the Badge page shows `v0.9.3`.